

Finding Hidden Functional States from EEG Based on Topological Features.

Supplementary materials B.

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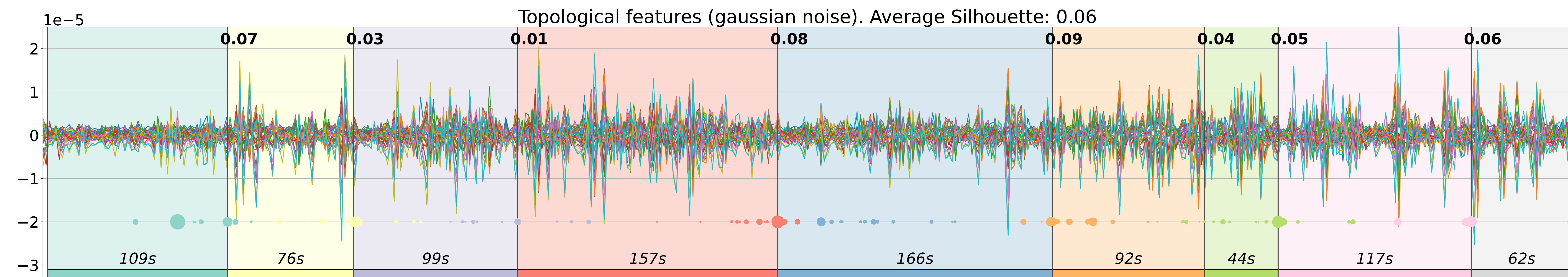
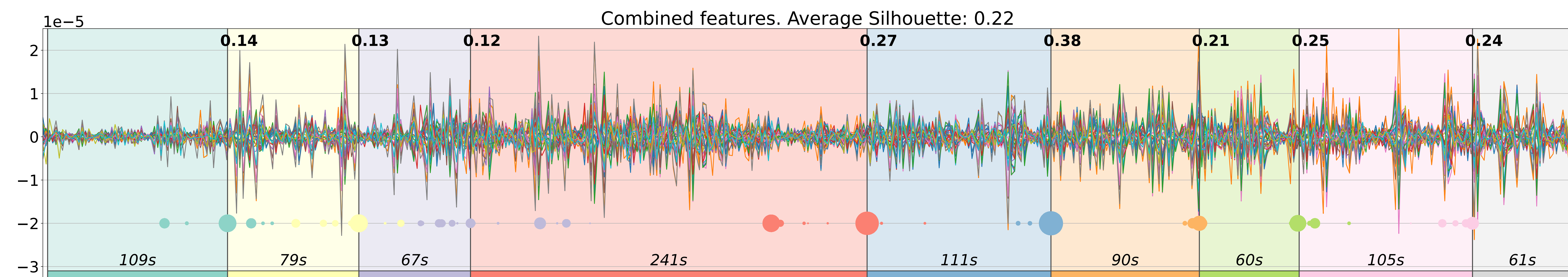
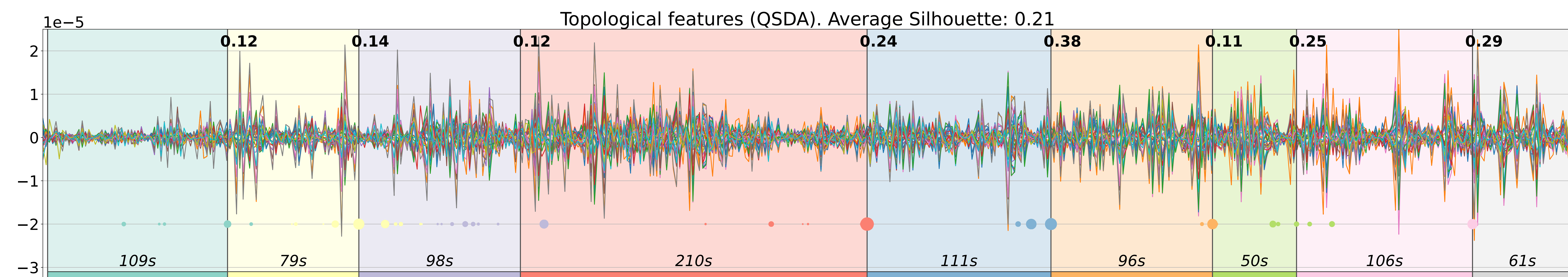
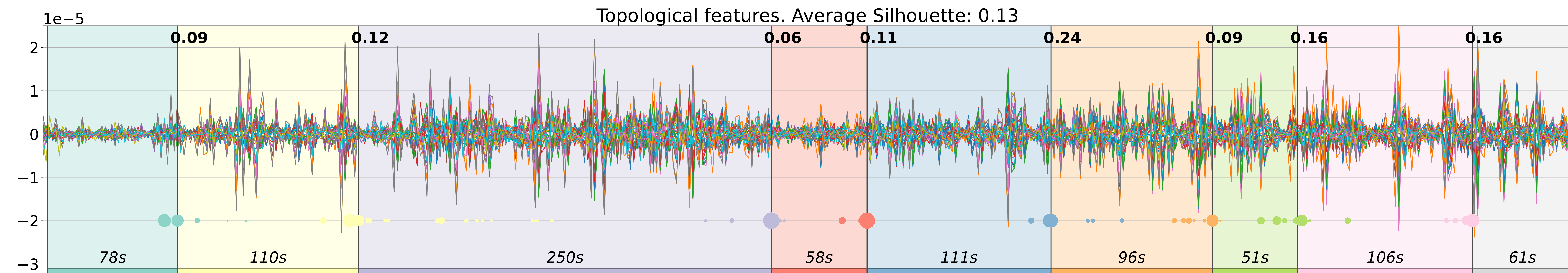
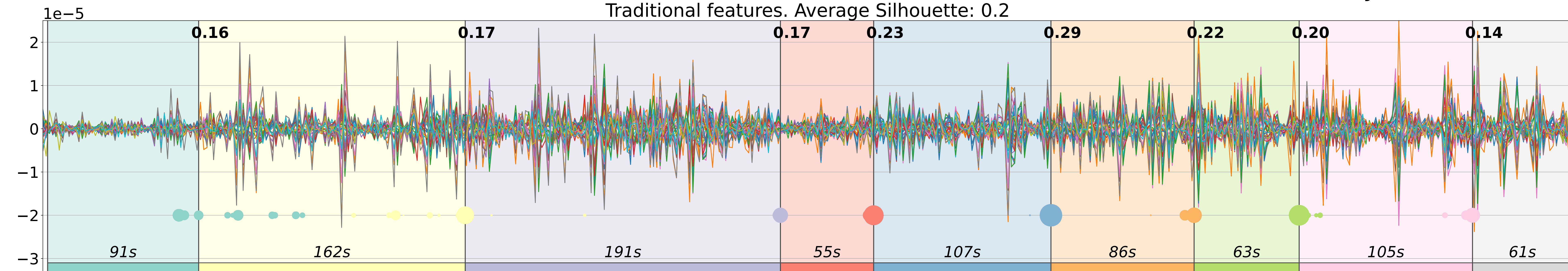
³ Lomonosov Moscow State University, Moscow, Russia

1. Supplementary materials. Result with full feature set for subjects 1 – 3.

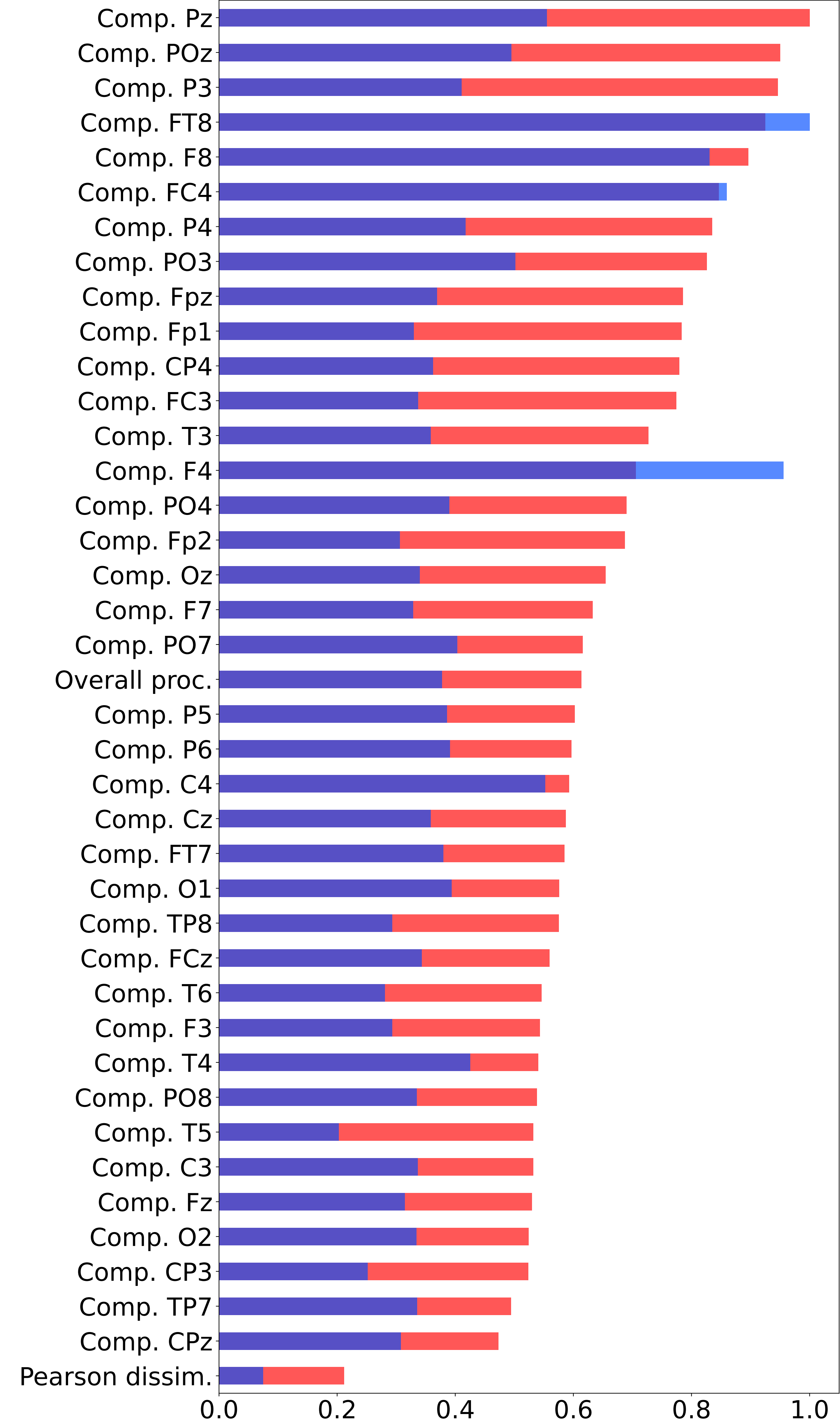
The left columns contain the EEG recordings of respective subjects with detected stages outlined in different colors and their durations indicated at the bottom of the diagrams. The charts display results obtained using all types of features and include the representations for noisy data. Vertical black lines indicate the final state boundaries computed by SDA. The multicolored scatter plots at the bottom of each chart represent the joint arrays of state boundary candidates obtained in the second step of SDA, with the dot size reflecting the frequency of occurrence of an epoch in this array and illustrating the stability of the boundary. The black numbers at the top of each chart present the silhouette coefficients for pairs of adjacent stages of each discovered boundary.

The bar charts in the remaining columns display mean information values (blue) and QSDA scores (red) of all topological features for each subject, aggregated by the parameters of their source: extraction technique and brain region, vectorization method, simplex dimension, and statistical characteristic. At the top right corner of each figure, we present heatmaps of feature quality scores aggregated by the sensor position during the experiment.

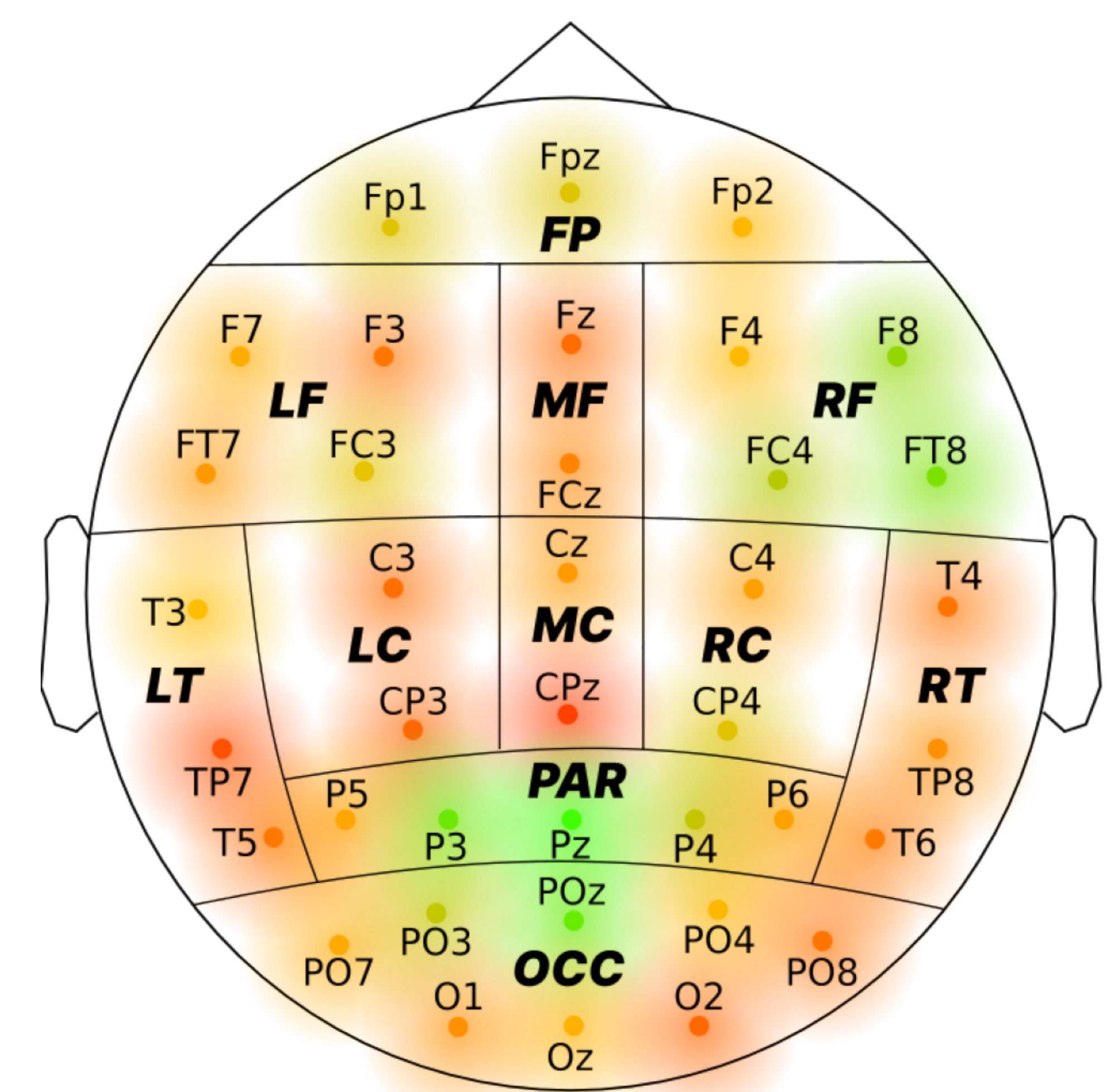
Subject 1



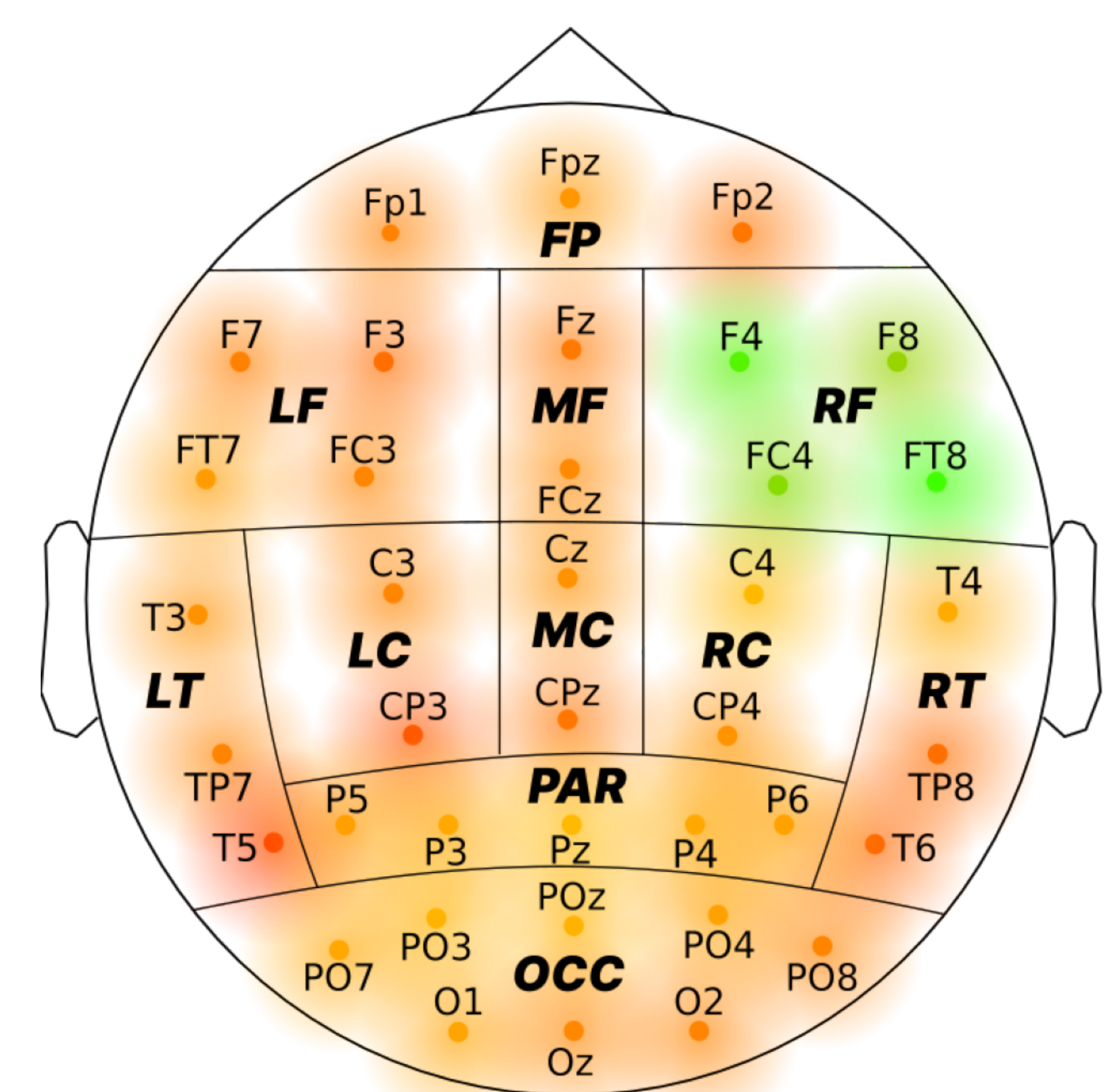
Extraction techniques and brain regions



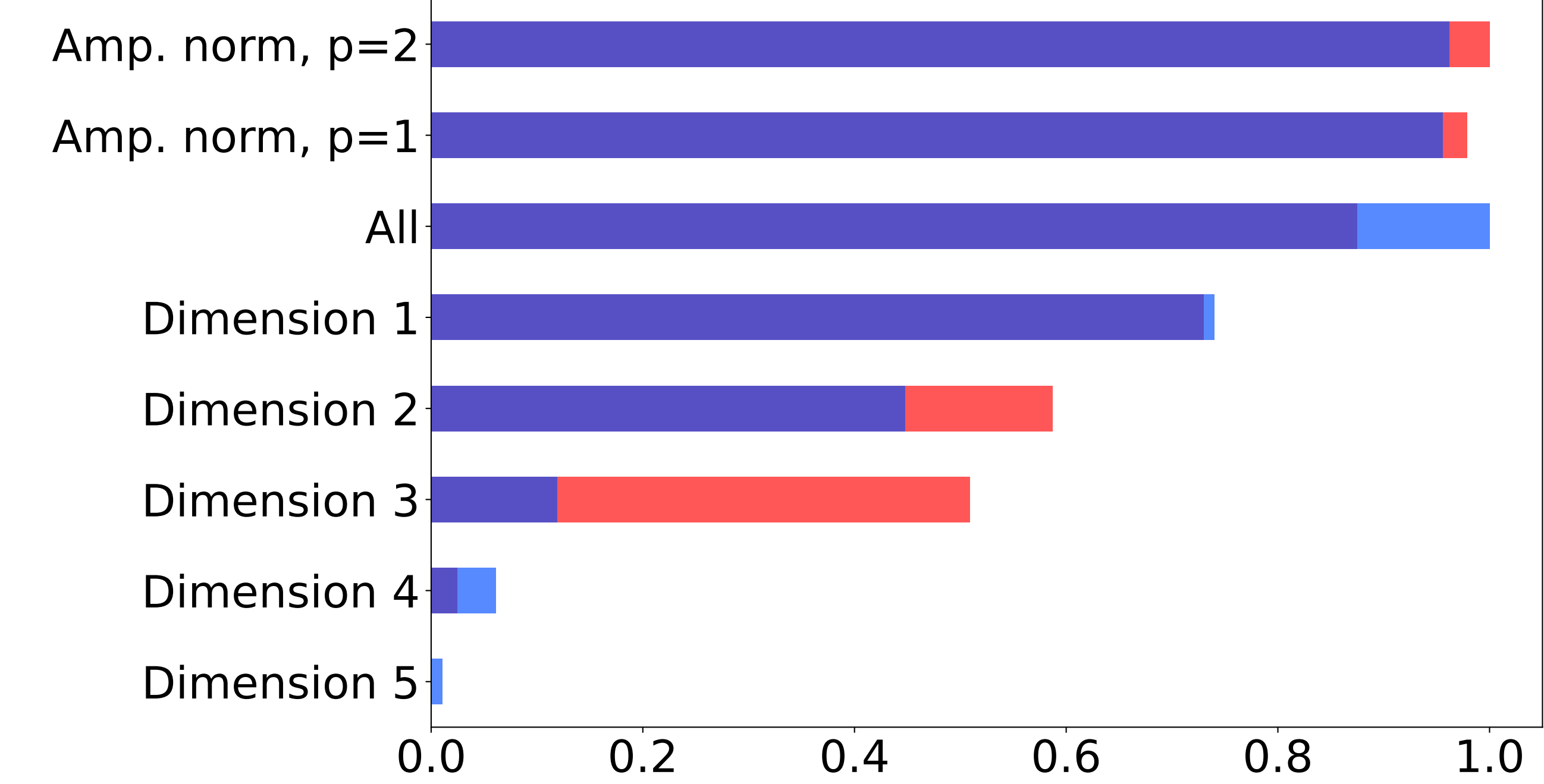
QSDA scores



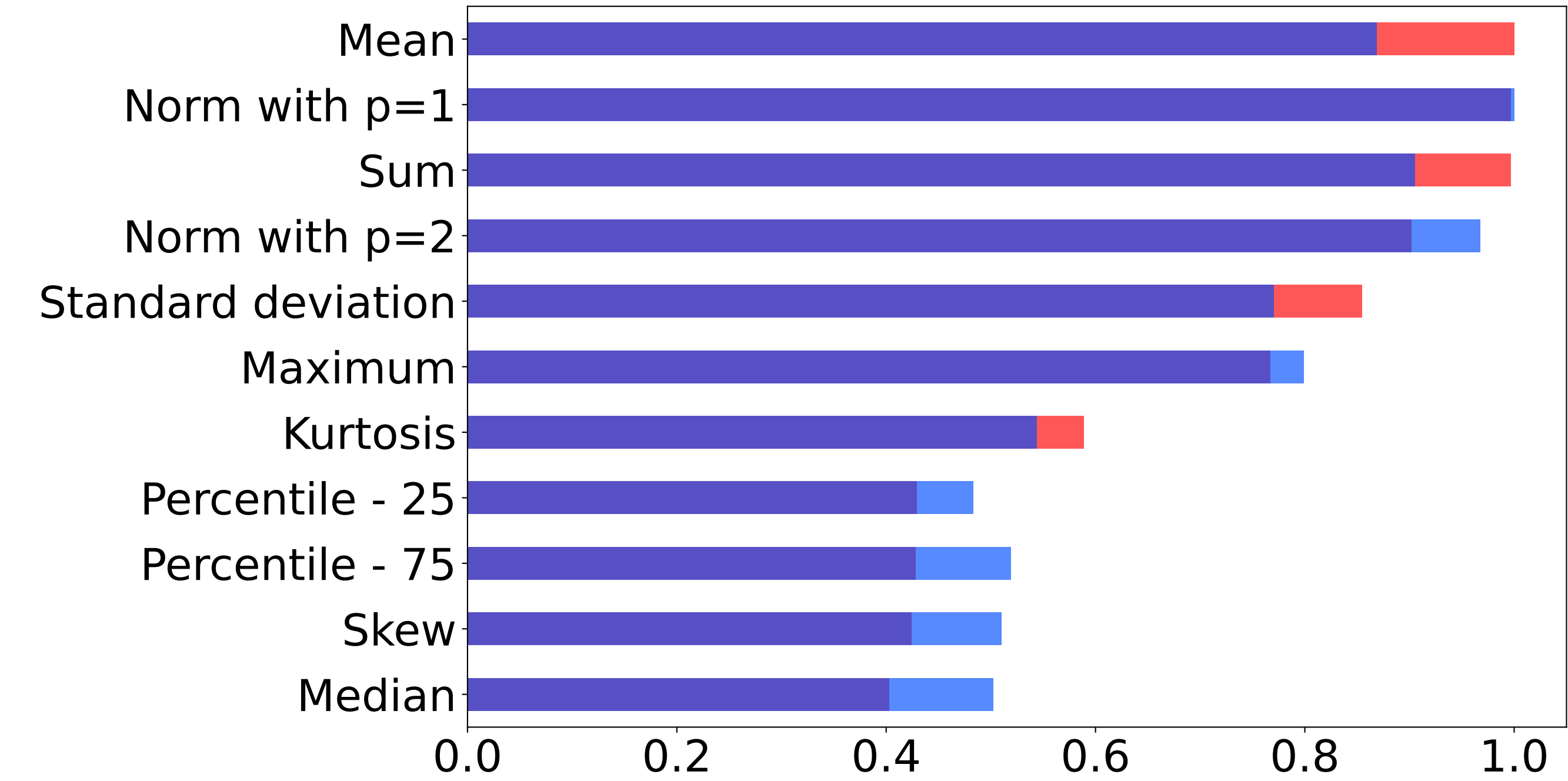
IV scores



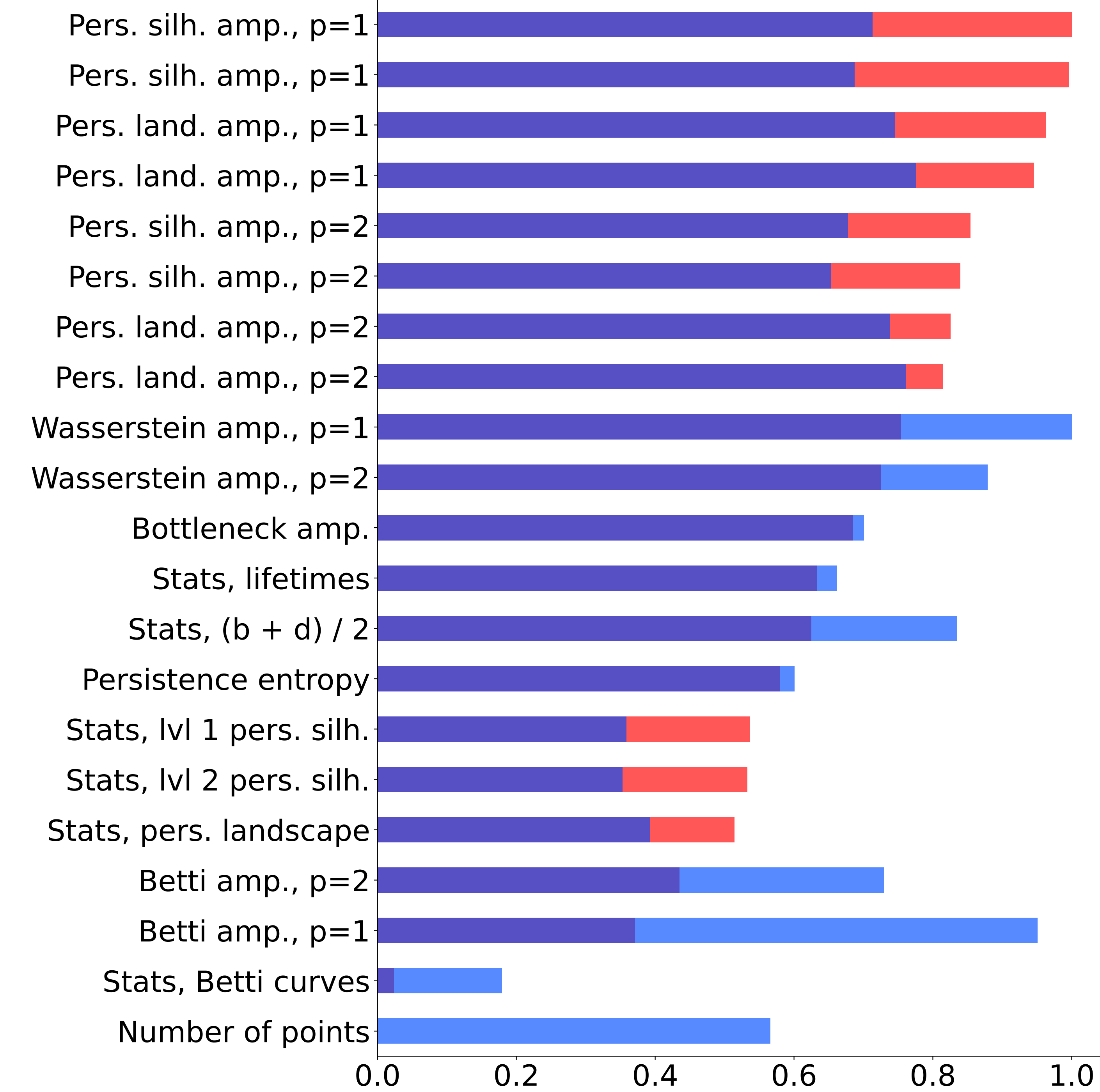
Simplex dimensions



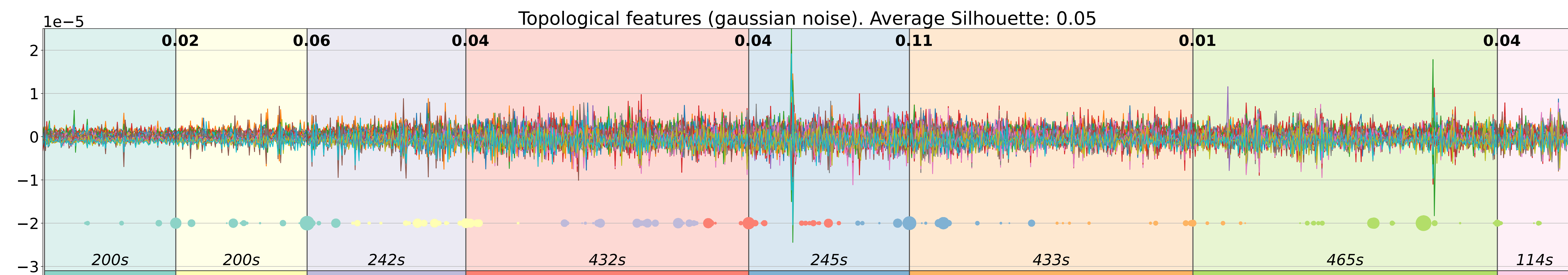
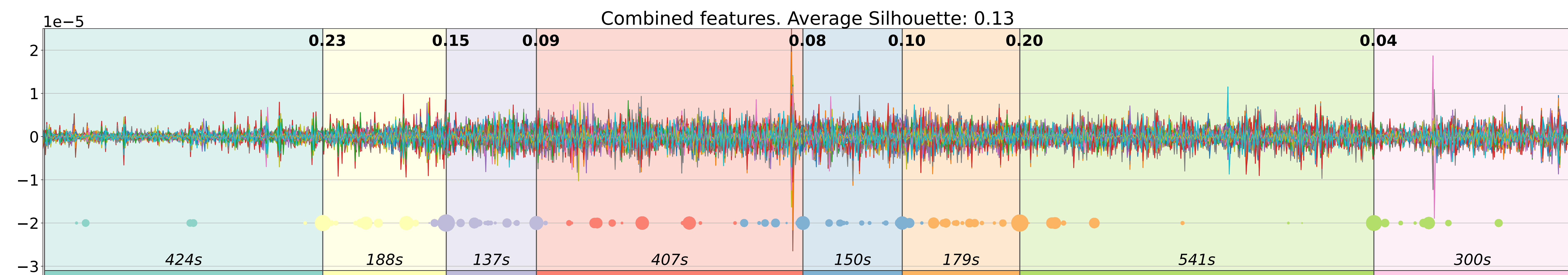
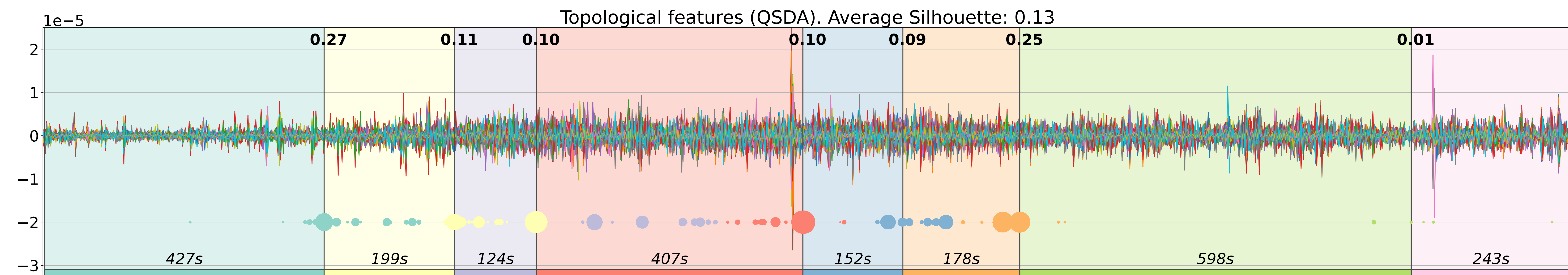
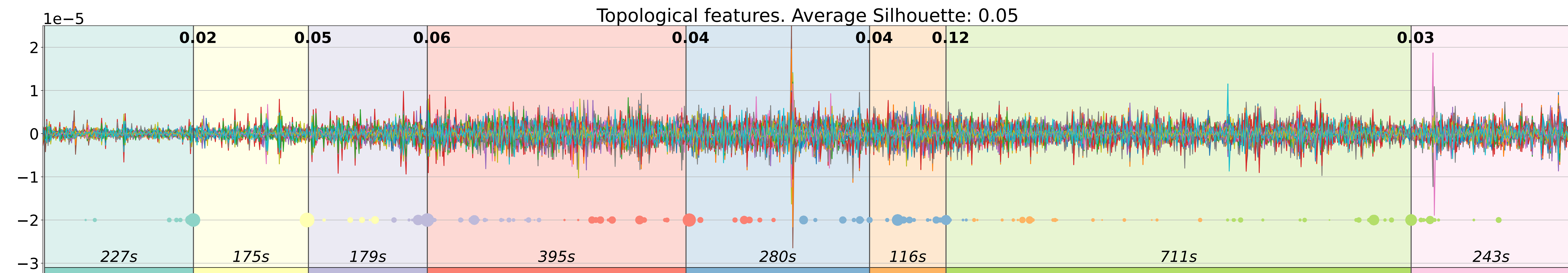
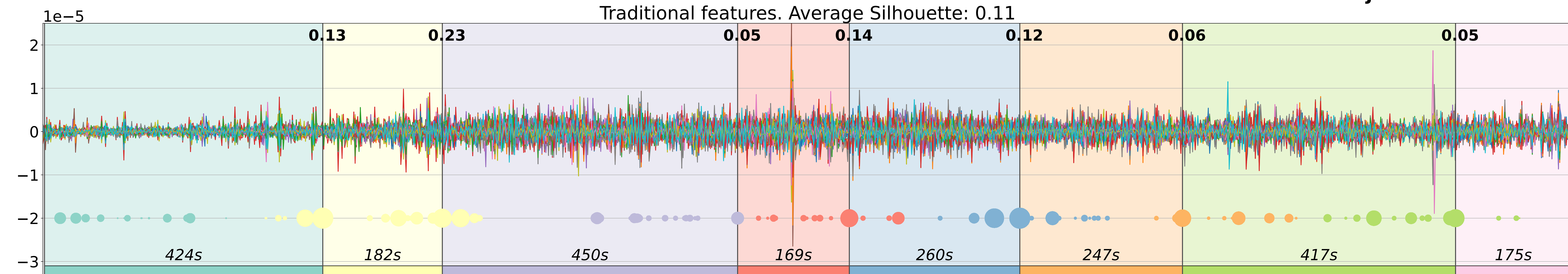
Statistical characteristics



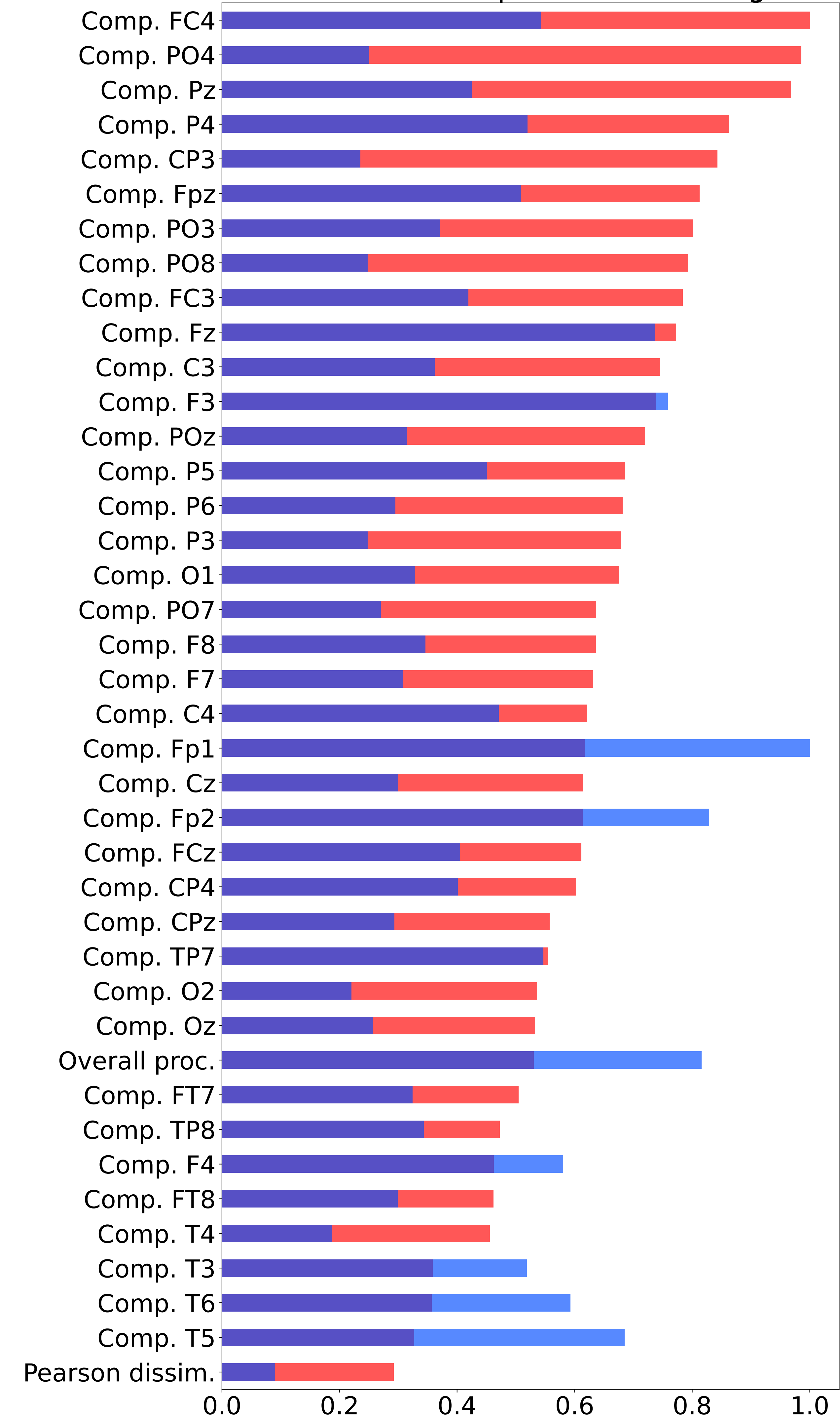
Vectorization methods



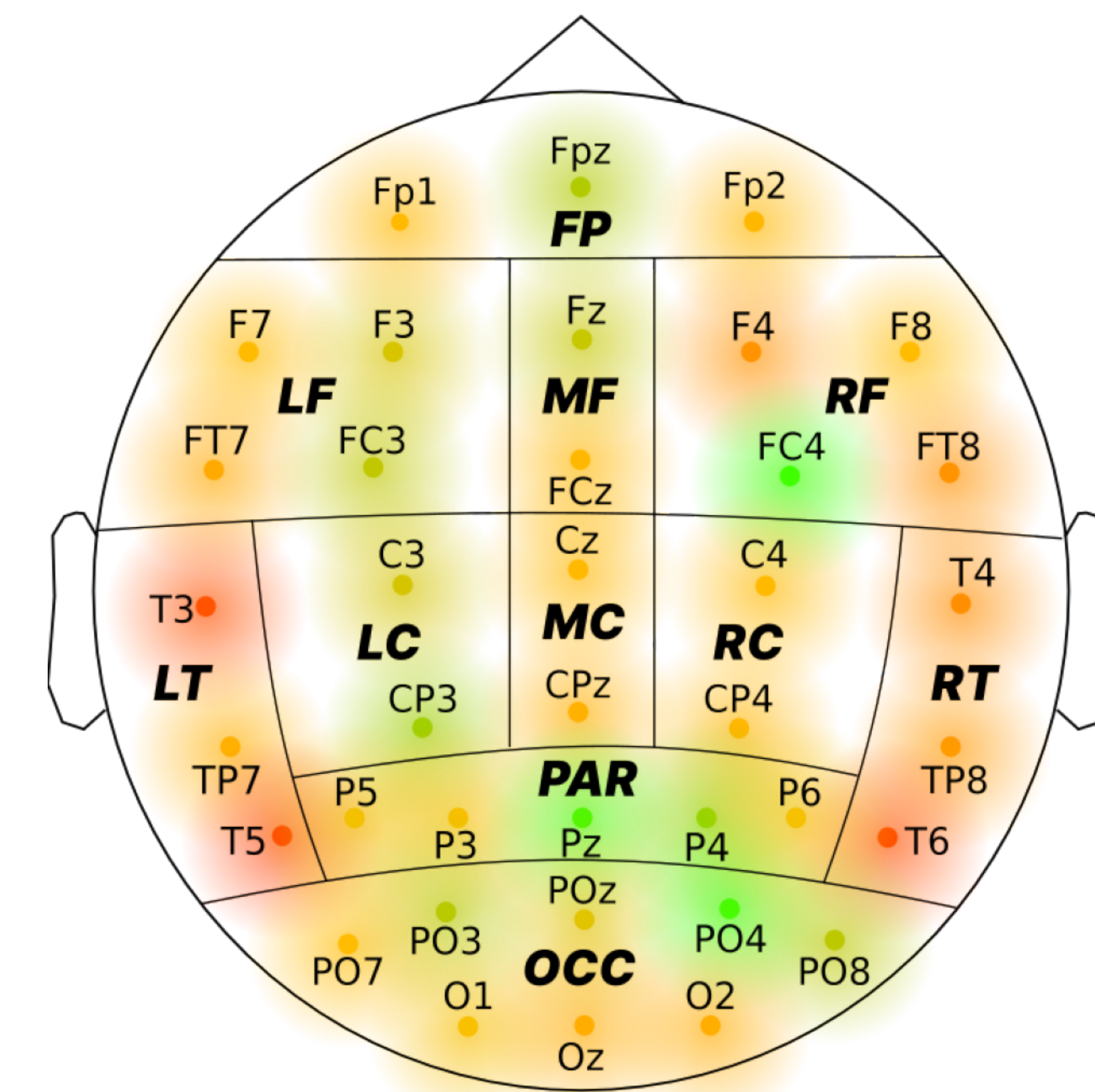
Subject 2



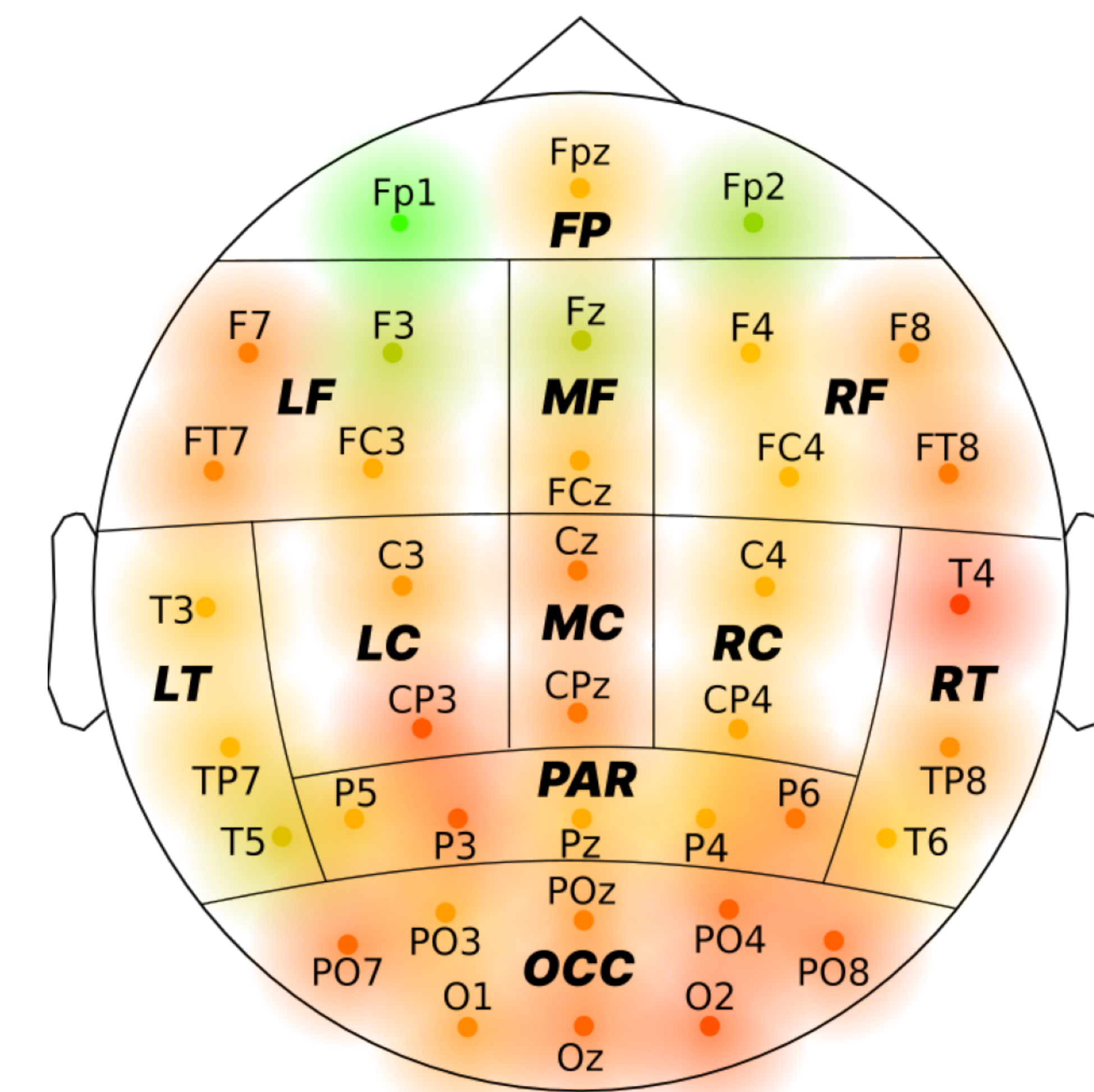
Extraction techniques and brain regions



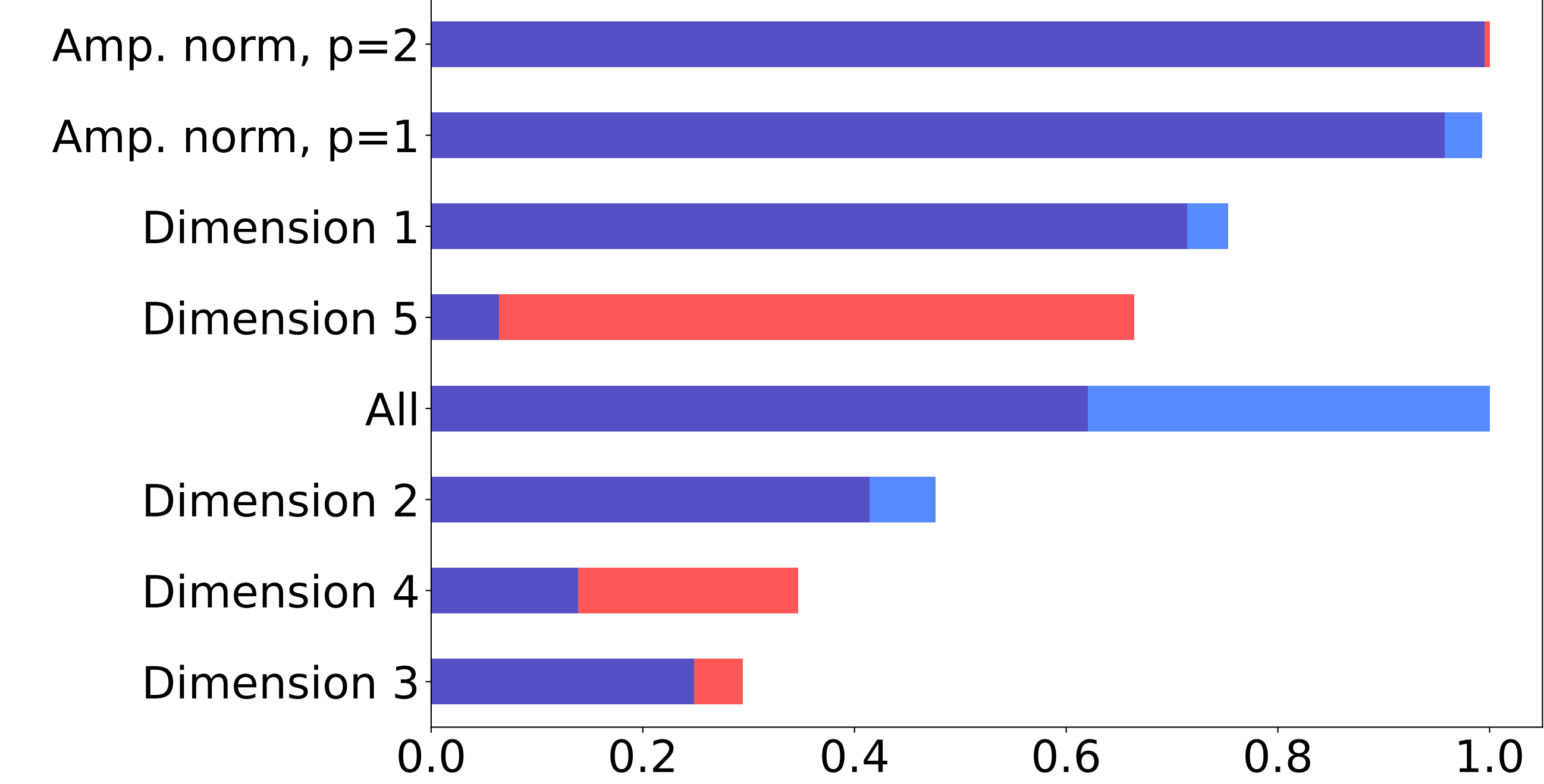
QSDA scores



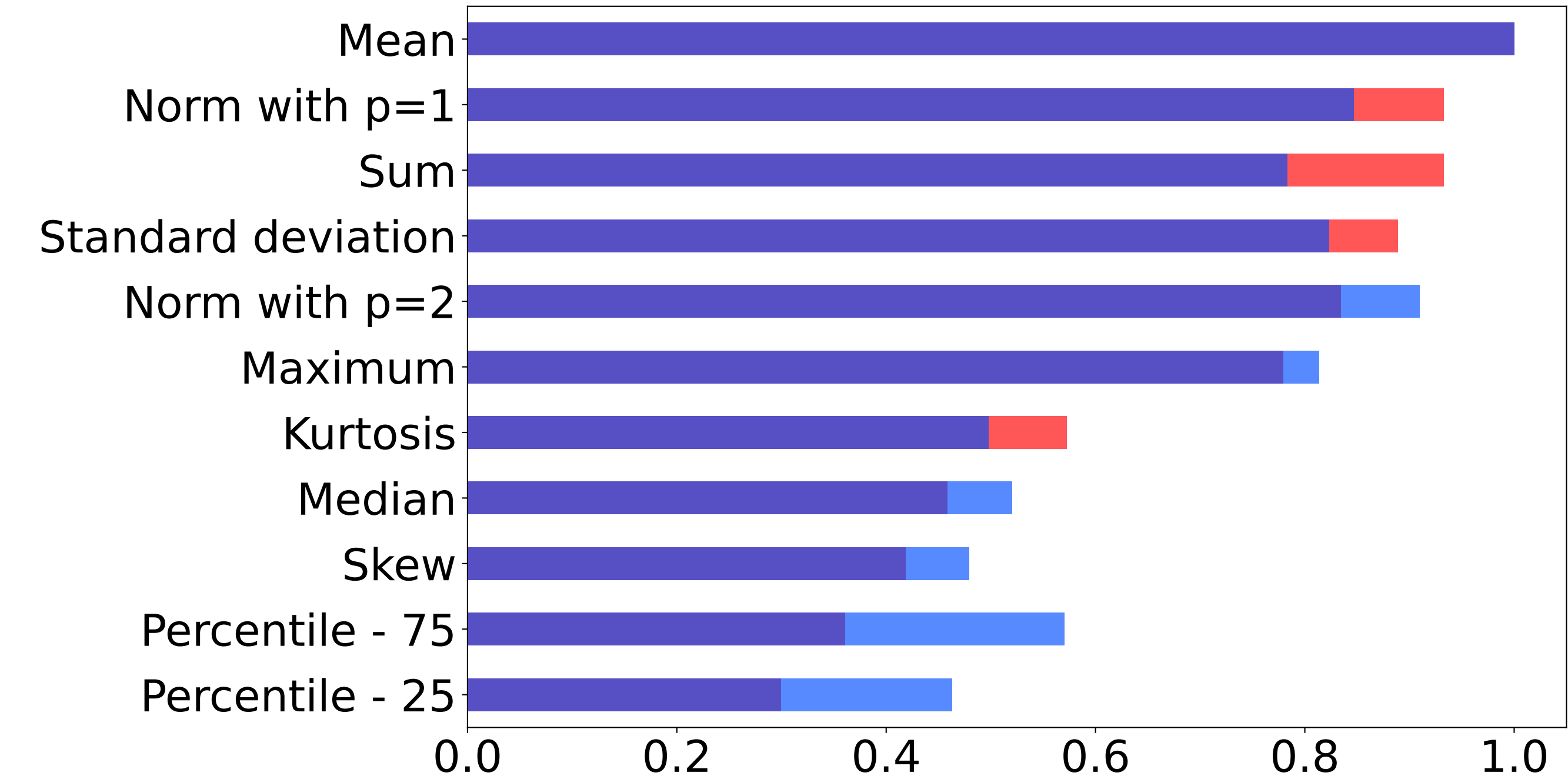
IV scores



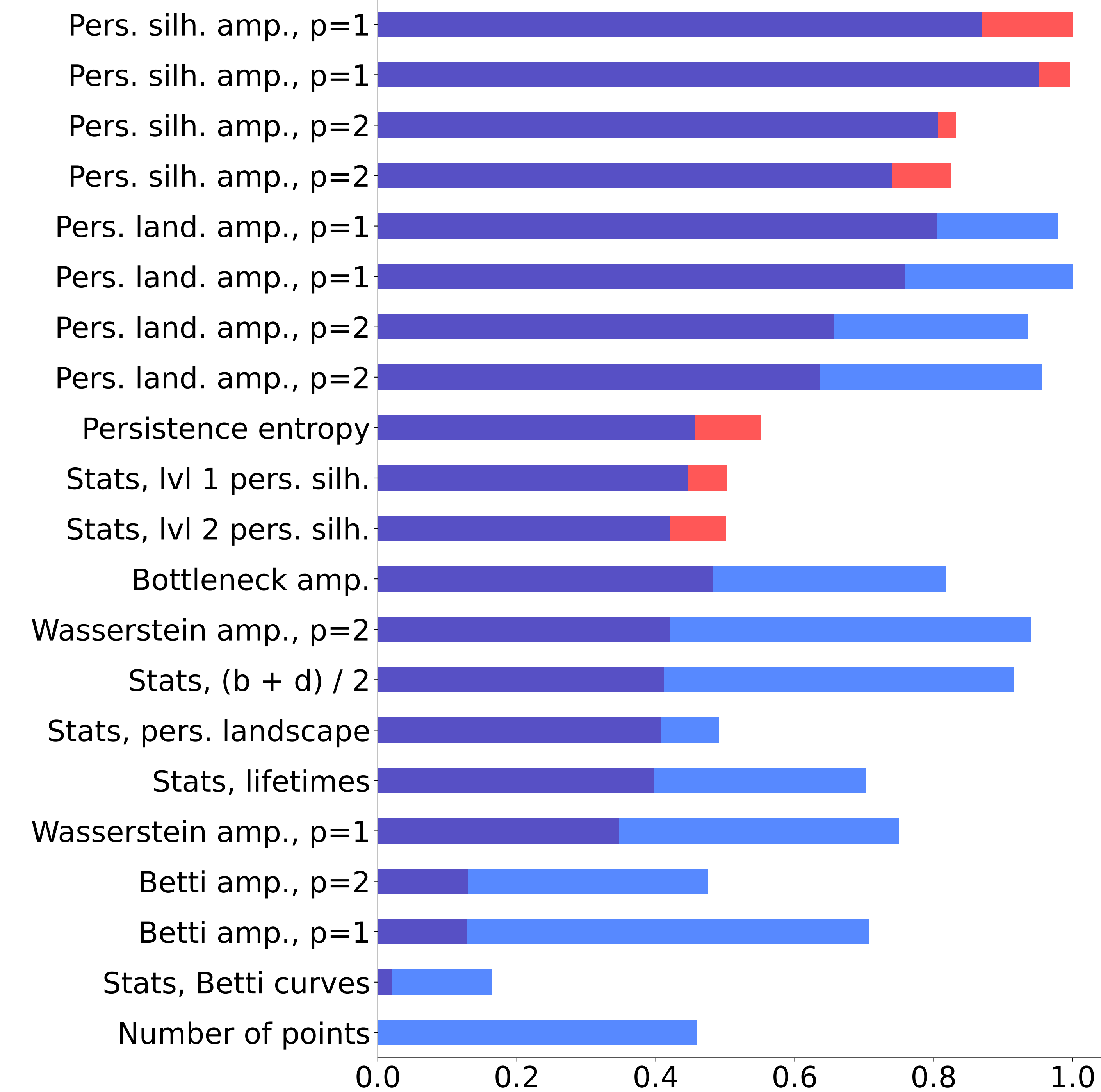
Simplex dimensions



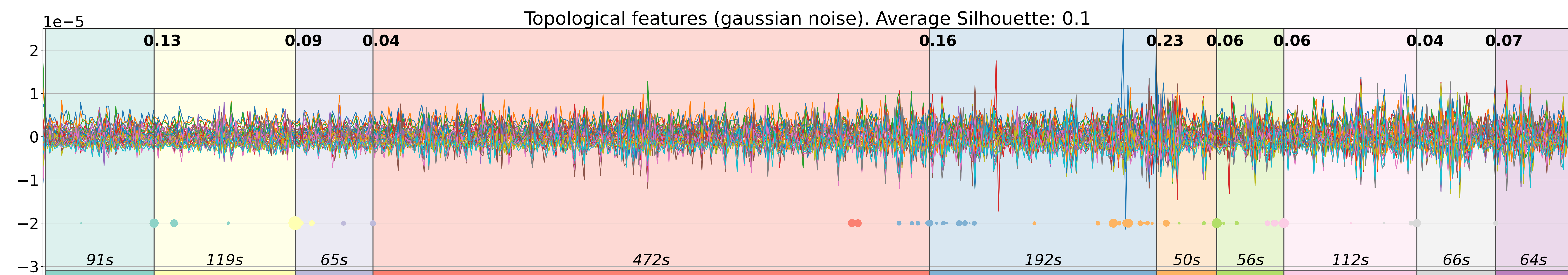
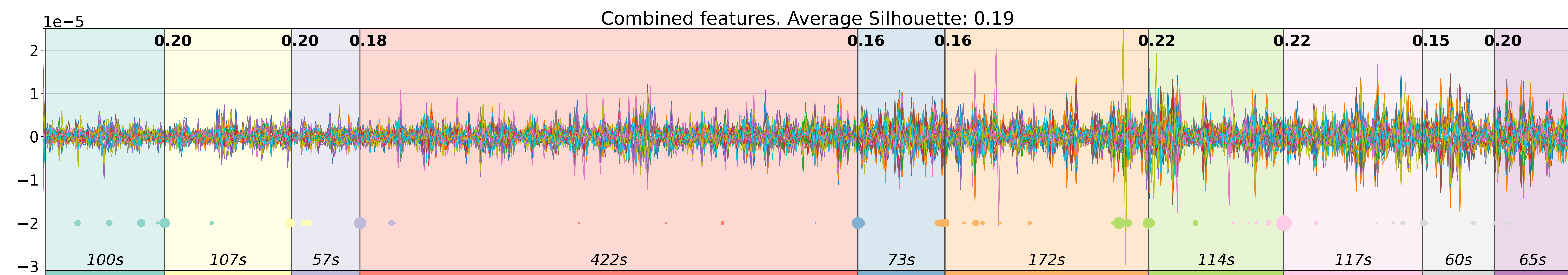
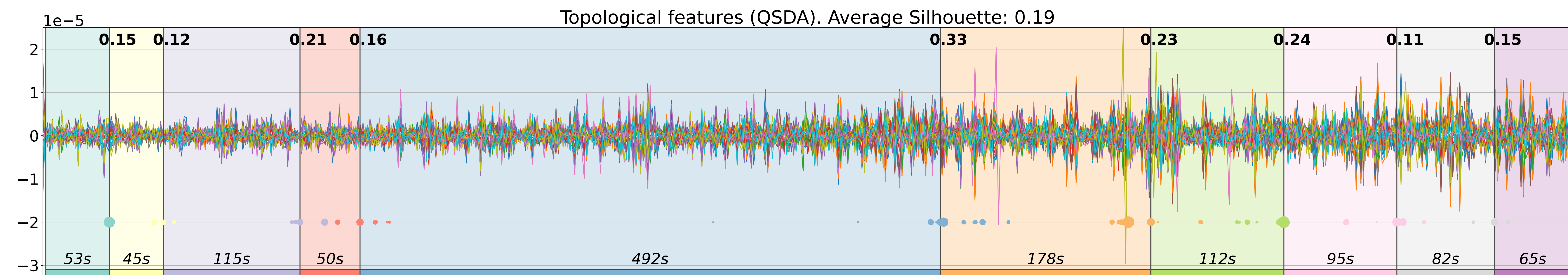
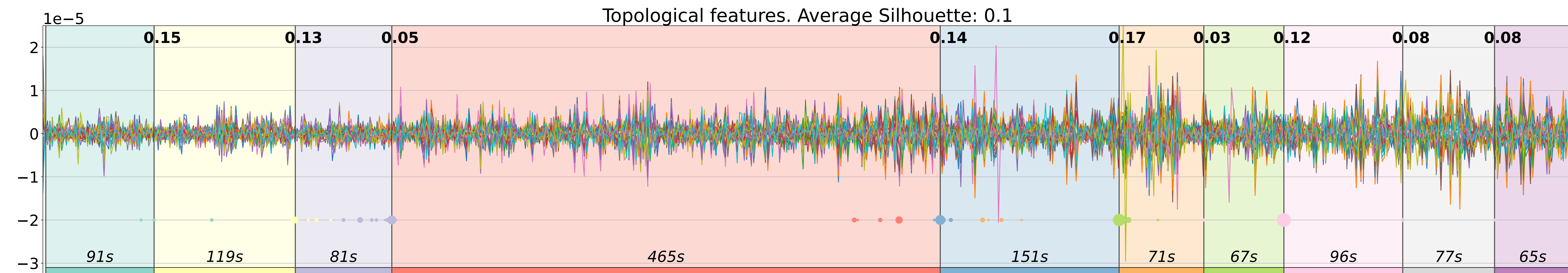
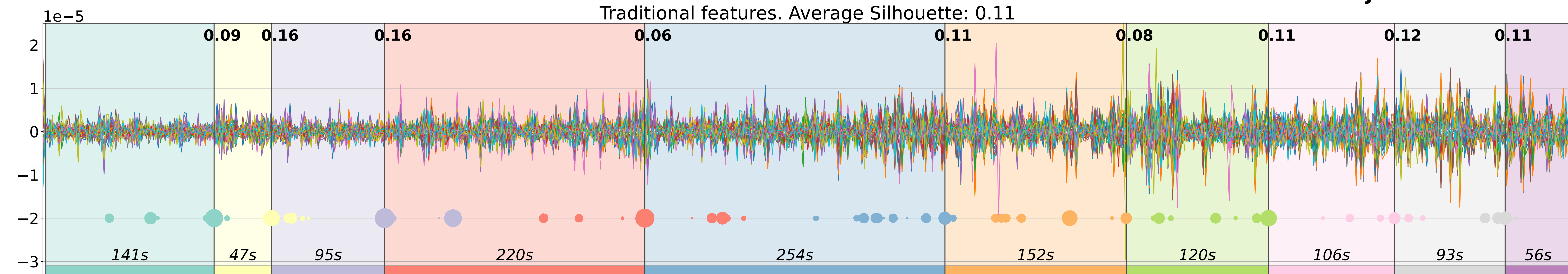
Statistical characteristics



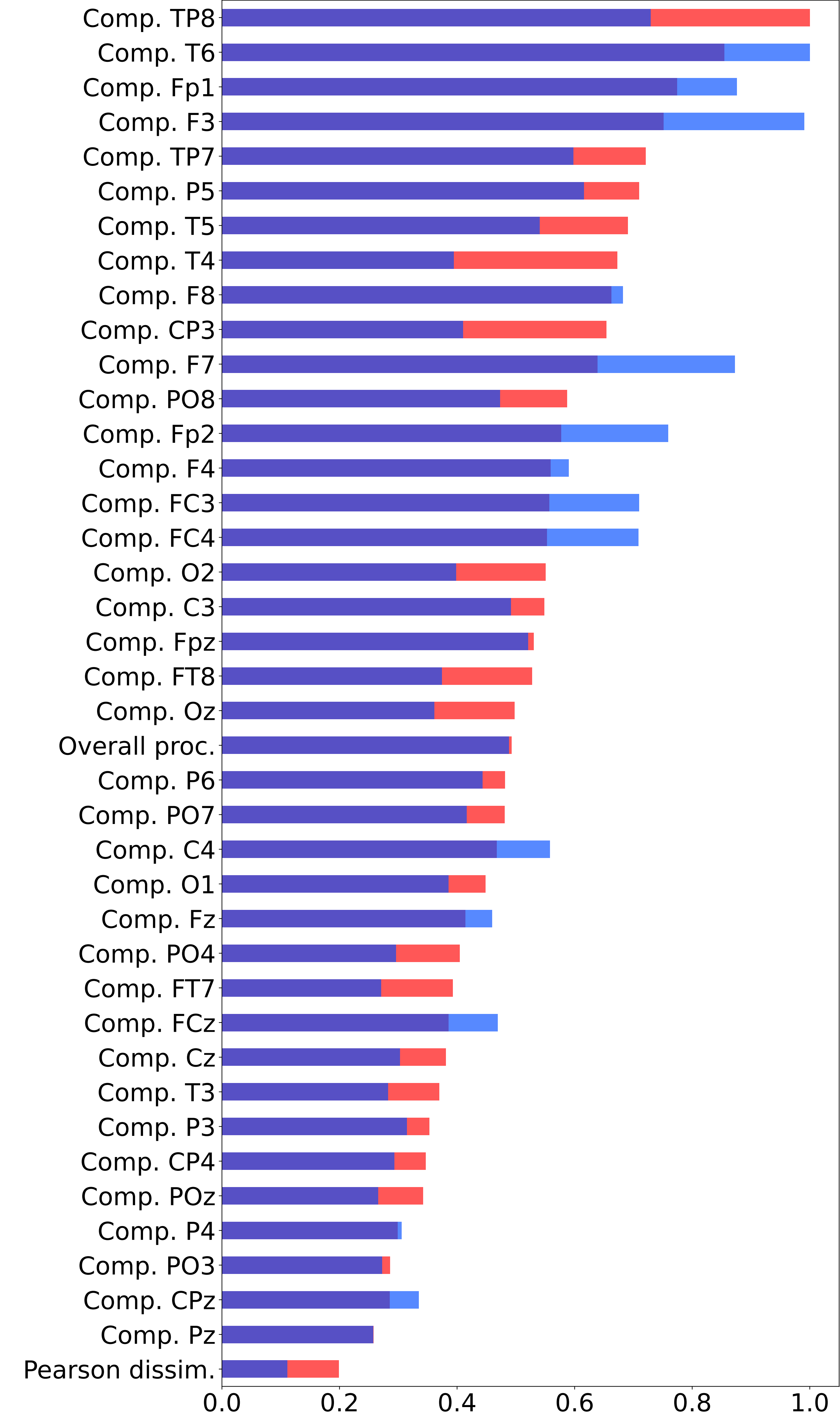
Vectorization methods



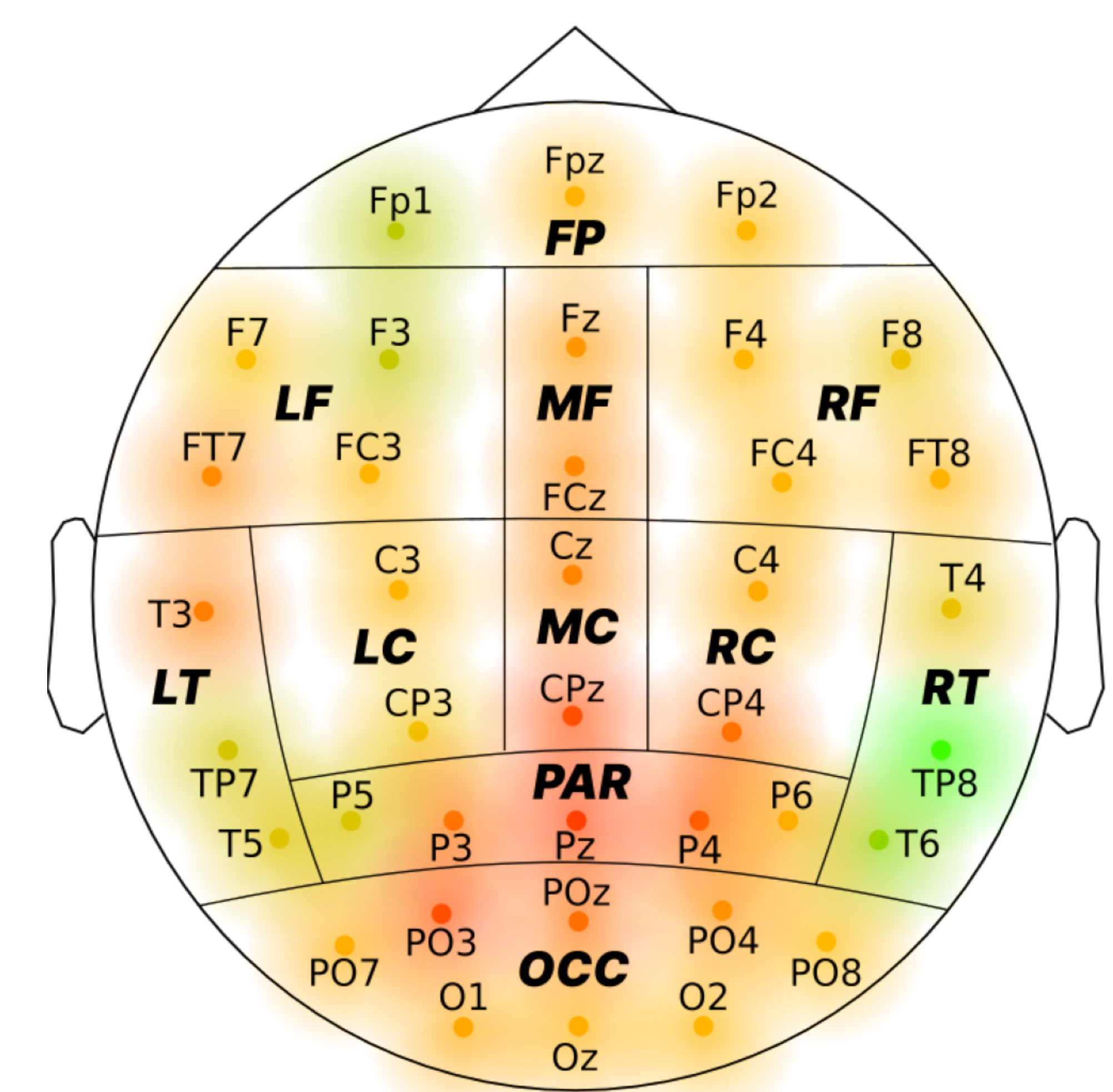
Subject 3



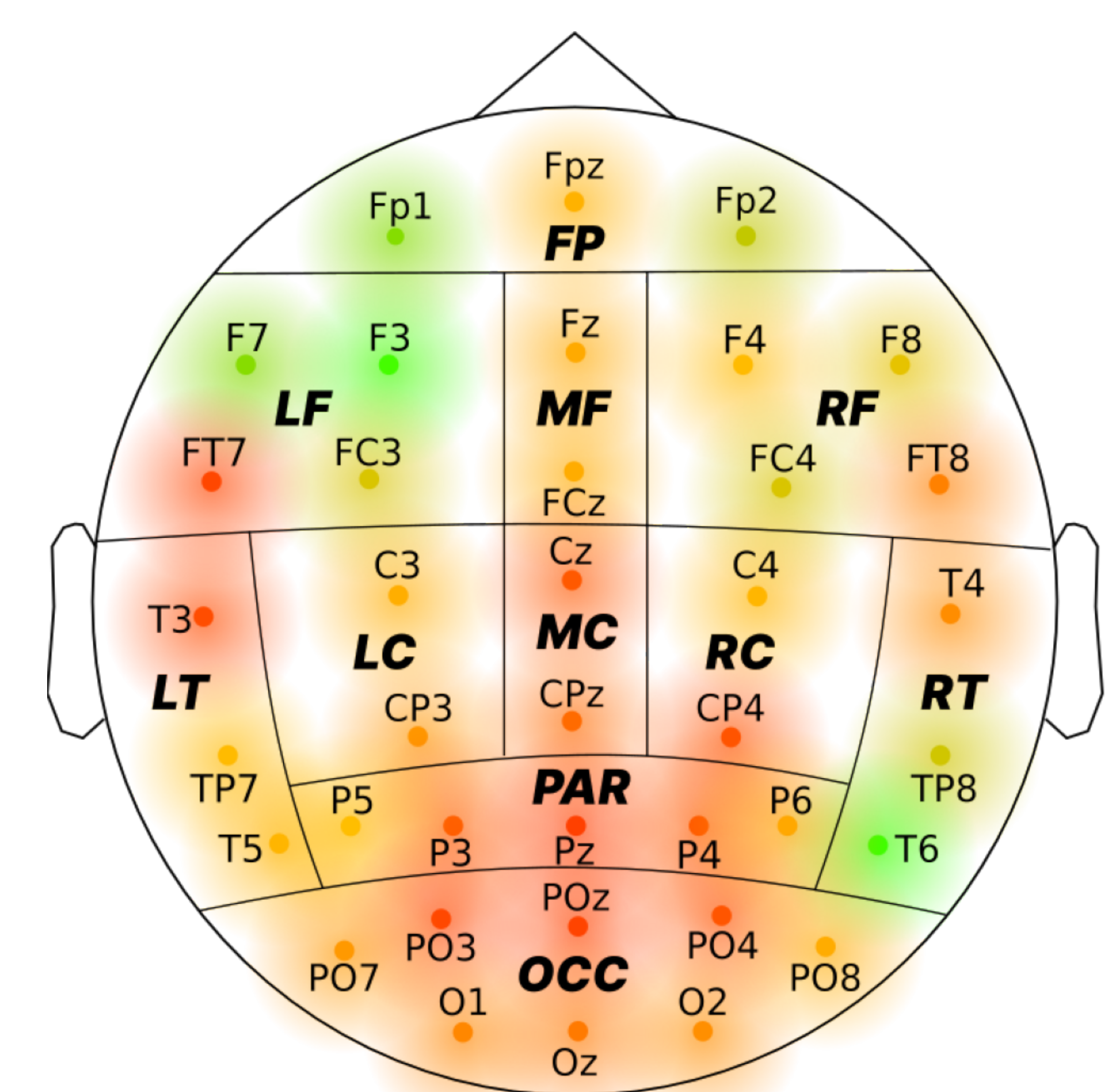
Extraction techniques and brain regions



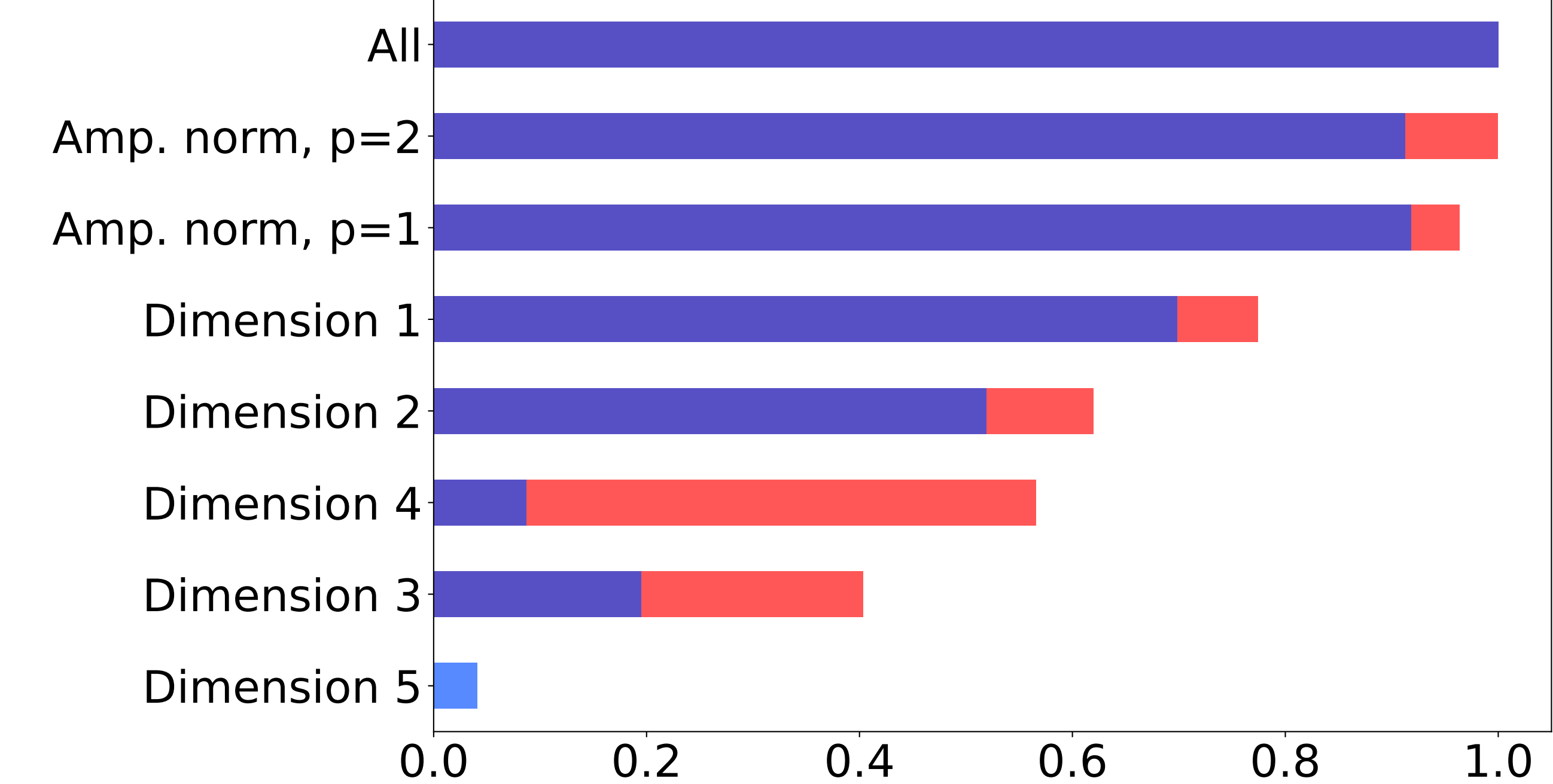
QSDA scores



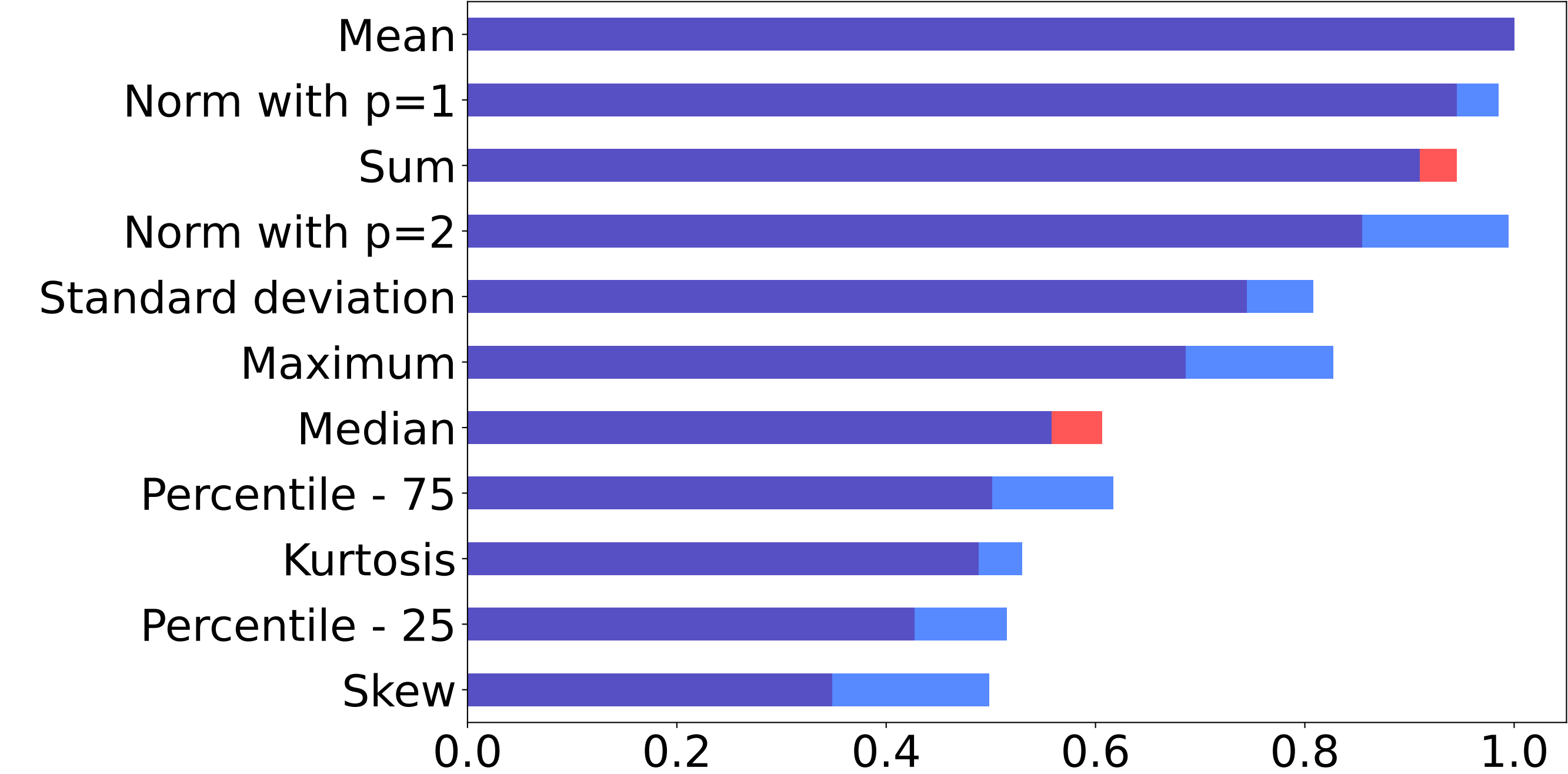
IV scores



Simplex dimensions



Statistical characteristics



Vectorization methods

